

Selective Repeat Protocol (SRP): Technical Briefing

Summary

Selective Repeat Protocol (SRP), also called Selective Repeat ARQ, is an advanced sliding window protocol used for reliable data transfer in the Data Link and Transport layers.

Unlike Go-Back-N (GBN), SRP retransmits **only the specific lost or corrupted frames**, significantly improving bandwidth efficiency.

Key characteristics:

- Selective retransmission of lost frames
- Symmetrical sender and receiver windows
- Out-of-order buffering at the receiver
- Individual timers for each frame
- Higher channel utilization compared to GBN

1. Core Principles of Selective Repeat

1.1 Window Management

In SRP:

- Sender Window (W_s) > 1
- Receiver Window (W_r) > 1
- Typically: $W_s = W_r$

Because the receiver accepts out-of-order packets:

- It must buffer frames that arrive within its window.
- Packets are delivered to the upper layer strictly in order.

This requires more memory and logic compared to GBN.

1.2 Acknowledgment Strategy

SRP uses **independent acknowledgments**, unlike GBN's cumulative ACKs.

- ACK number corresponds directly to the frame number received.
- If Frame 0 is received → ACK 0 is sent.
- One-to-one mapping between frame number and ACK number.

This allows precise retransmission of only missing frames.

1.3 Timer and Search Logic

Individual Timers

- Each frame in the sender window has its own timer.
- If a timer expires, only that frame is retransmitted.

Sender Searching

- Sender must search its buffer to locate and retransmit the specific frame.

Receiver Sorting

- Receiver stores out-of-order packets.
- Frames are sorted before in-order delivery to upper layers.

This increases computational complexity compared to GBN.

2. Operational Dynamics

2.1 Handling Out-of-Order Frames

- Receiver accepts packets within its window range.
- Out-of-order frames are buffered.
- Delivery to upper layers occurs only after missing frames arrive.

2.2 Negative Acknowledgments (NAK)

SRP may use NAK to speed up recovery.

Mechanism:

1. If Frame 3 is missing but Frame 4 arrives:
 - Receiver sends ACK 4.

- Receiver sends NAK 3.
- 2. Upon receiving NAK 3:
 - Sender immediately retransmits Frame 3.
 - No need to wait for timeout.

NAK Limitation

- Typically sent only for the first missing packet.
- Prevents excessive control traffic.
- If NAK is lost, the frame's timer eventually expires.

2.3 Window Freezing

If the earliest unacknowledged packet is missing:

- Sender window cannot slide forward.
- Receiver cannot deliver buffered packets beyond the gap.
- System resumes normal sliding only after the missing packet is received.

3. Mathematical Framework

3.1 Sequence Number Constraint

To avoid ambiguity between new and old frames:

$$W_s + W_r \leq \text{Available Sequence Numbers (ASN)}$$

For the common case $W_s = W_r$:

$$2W_s \leq \text{ASN}$$

If k bits are used:

$$\text{ASN} = 2^k$$

Therefore:

$$\text{Maximum } W_s = 2^{k-1}$$

Minimum sequence numbers required:

$$2W_s$$

This is stricter than GBN, which requires only $N + 1$ sequence numbers.

4. Efficiency and Throughput

Let:

T_t = Transmission delay

T_p = Propagation delay

$a = T_p / T_t$

4.1 Efficiency

$$\eta = (W_s \times T_t) / (T_t + 2T_p)$$

If divided by T_t :

$$\eta = W_s / (1 + 2a)$$

4.2 Throughput

Throughput = $\eta \times \text{Bandwidth}$

OR

Throughput = $(W_s \times \text{Frame Size}) / \text{Total Time}$

SRP achieves maximum efficiency when:

$$W_s \geq 1 + 2a$$

This matches the Bandwidth-Delay Product requirement.

5. Comparative Analysis

Feature	Stop-and-Wait	Go-Back-N (GBN)	Selective Repeat (SR)
Efficiency	$1 / (1 + 2a)$	$N / (1 + 2a)$	$W_s / (1 + 2a)$

Sender Window	1	N	N
Receiver Window	1	1	N
ACK Type	Independent	Cumulative	Independent
Retransmission	Single frame	Entire window	Single frame
Receiver Buffer	1	1	N
Sequence Numbers	2	$N + 1$	$2N$
Max W_s (k bits)	1	$2^k - 1$	2^{k-1}

6. Technical Case Study: Satellite Link Example

Given:

- Link speed = 1 Mbps
- Satellite altitude = 36,504 km
- Signal speed = 3×10^8 m/s
- Efficiency = 25%
- Protocol = GBN-127

Step 1: Propagation Delay

Ground → Satellite → Ground

Distance:

$$2 \times 36,504 \text{ km} = 73,008 \text{ km}$$

Velocity:

$$3 \times 10^5 \text{ km/s}$$

T_p :

$$73,008 / 300,000 = 0.24336 \text{ s}$$

Step 2: Apply Efficiency Formula

$$\eta = (n \times T_t) / (T_t + 2T_p)$$

Given $\eta = 1/4$ and $n = 127$:

$$1/4 = (127 \times T_t) / (T_t + 2 \times 0.24336)$$

Solving:

$$T_t \approx 0.00096 \text{ s}$$

Step 3: Packet Size

$$\text{Frame Size} = \text{Bandwidth} \times T_t$$

$$= 10^6 \times 0.00096$$

$$= 960 \text{ bits}$$

$$= 120 \text{ Bytes}$$